Immanuel Tampubolon



AI Engineer Intern · LLM agents, RAG and multi-agent systems

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Seeking an internship. Available onsite, including relocation abroad, or remote.

PROFILE

Computer Technology student at Institut Teknologi Del. I build AI systems that have to work on real input, not on a demo prompt. A coding agent that patches repositories it has never seen and proves each fix by running that project's own test suite in a sandbox, validated against the open-source Flask codebase. A RAG security assistant grounded in MITRE ATT&CK and a custom corpus of Indonesian scam cases, scored with RAGAS, rated 4.3 of 5 by 16 test users, and now the basis of my thesis. An anomaly-detection pipeline that keeps deciding when the model API is throttled, because an ensemble of Isolation Forest and Random Forest carries the volume and the model sees only high-risk traffic. What interests me most is what decides whether an AI feature survives production: retrieval quality, and what happens when the model is not there. I am looking for an AI Engineer internship where I can build these alongside a team.

TECHNICAL SKILLS

LLM and agents	RAG pipelines, multi-agent orchestration, tool and function calling, self-correction loops, prompt engineering, LangChain
Model integration	Provider-agnostic LLM APIs (Groq, Hugging Face, Gemini, Cohere), response streaming over Server-Sent Events, rate-limit fallback and circuit breakers
Retrieval and ML	Embeddings (Cohere, sentence-transformers), Supabase pgvector, cosine-similarity search, scikit-learn (Isolation Forest, Random Forest)
Evaluation	RAGAS answer-quality scoring, Langfuse run-level tracing
Languages	Python, TypeScript, JavaScript, Go, PHP. Foundational Java and C
Backend and frontend	FastAPI, Flask, Node.js, Laravel, Go. React, Next.js, React Native, Tailwind CSS
Data and DevOps	PostgreSQL, MySQL, MongoDB, Redis Streams, Docker, Nginx, Git, Vercel, Railway, Render

AI ENGINEERING PROJECTS

Sentinel. Autonomous self-correcting coding agent

Personal project, 2026

Python, FastAPI, Llama 3.1 via Hugging Face Inference Providers, sentence-transformers, Docker, Langfuse, Next.js

- The model proposes a patch, the suite runs in an isolated sandbox — Docker, or a virtualenv provisioned per project — and failing test output goes back to the model for up to three further attempts.
- Every fix is grounded in repository context through a sentence-transformers retriever ranking by cosine similarity, so the model reasons over related modules instead of the target file alone.
- Takes any external repository by GitHub clone or ZIP upload and resolves its real dependencies automatically, with no per-project configuration.
- Run-level tracing goes to Langfuse, and a Next.js interface streams the agent's reasoning over Server-Sent Events as it retrieves, generates and runs tests.

AnomalyGuard. Multi-agent network anomaly detection

Personal project, 2026

Python, Flask, Redis Streams, Groq LLM with function calling, scikit-learn, Next.js, Recharts

- Three agents — detection, decision, response — communicate over Redis Streams. The investigating agent works through function-calling tools for incident history, active-incident count and traffic baseline.
- The circuit breaker drops back to rule-based decisions when the API is unavailable. Keeping the system useful under quota limits turned out to be most of the work.
- A Next.js and Recharts dashboard pulls live data from the three backend services across eight views, including LLM quota health and the dead-letter queue.

CyberGuard AI. RAG security assistant for Indonesian users

Team of three, basis of thesis

FastAPI, LangChain, Cohere embeddings, Qwen, Supabase pgvector, RAGAS, Langfuse, Next.js, Railway

- Implemented the retrieval pipeline end to end — ingestion, Cohere embeddings, pgvector storage, LangChain retrieval and generation — across a corpus of 1,292 documents, 50 or more of them scam case studies compiled for the project.

- Added multi-turn conversational memory and reworked retrieval where sources overlapped, with every request traced in Langfuse. Measured with RAGAS over a 19-question cybersecurity set and a 16-respondent user study that averaged 4.3 of 5 on usability, answer quality and performance.
- Also shipped a malicious-URL verification tool and an interactive security learning module, with the FastAPI backend deployed on Railway.
- For my thesis I am splitting it into separate retriever, verifier and responder agents, with LoRA fine-tuning for Indonesian-language digital-fraud detection.

DevPulse. AI code review SaaS *Personal project*

Go, Next.js, Groq (Llama 3.3), GitHub OAuth, Vercel, Railway

- Reviews GitHub pull requests automatically with Llama 3.3, flagging exposed secrets and other security leaks, and writes its reports in both English and Indonesian.
- Go backend with GitHub OAuth sign-in and a Next.js frontend, running in production on Vercel and Railway.

Apotek Century. Pharmacy mobile app with AI assistant *Full-stack project*

React Native (Expo), Laravel 11, FastAPI, Groq (Llama 3), Midtrans

- An AI medical assistant running on Llama 3 checks drug interactions inside a working pharmacy app, alongside real-time chat with a pharmacist.
- Developed the commerce side as well: catalogue with category filters and discounts, cart and Midtrans checkout, prescription upload, and an admin dashboard.

OTHER PROJECTS

Connect+, a five-service microservices campus platform deployed across three machines with a MongoDB replica set, Docker, Nginx and HAProxy. **Neo-Vanguard**, an AI-literacy and cybersecurity education platform in React and TypeScript. **Tanah Batak**, a cultural-heritage site in React, TypeScript and Go. **SiPerpus**, a library management system in Laravel. **UNTHOPEDINE**, a coffee-shop site with WhatsApp ordering.

EDUCATION AND ORGANISATION

Institut Teknologi Del, Diploma (D3, A.Md.) in Computer Technology *2024 to present, Toba*

Focus on software development, computer networking and embedded systems. Thesis in progress: developing CyberGuard AI into a modular multi-agent system with language-model fine-tuning for Indonesian-language digital fraud detection.

Himpunan Mahasiswa Teknologi Komputer, Facilities and Infrastructure Division *2025 to present*

CERTIFICATIONS, TRAINING AND COMPETITIONS

AI AND MACHINE LEARNING

- AI Engineer for Millennials, Komdigi DTS — *Feb 2026*
- AI Security Fundamentals, Komdigi and Microsoft DTS — *Feb 2026*
- Intro to Machine Learning, Kaggle — *Apr 2026*
- GenAI-Powered Data Analytics Job Simulation, Tata and Forage — *Sep 2026*

NETWORKING AND SECURITY

- Introduction to Cybersecurity, Cisco Networking Academy — *Feb 2026*
- HCIA-Datacom V1.0 Course, Huawei ICT Academy — *Nov 2025*
- HCIA-Storage V5.0 Course, Huawei ICT Academy — *Oct 2025*

SOFTWARE ENGINEERING

- Software Engineer Role Certification, HackerRank — *Sep 2026*
- Software Engineering Job Simulation, Commonwealth Bank and Forage — *Apr 2026*
- Memulai Pemrograman dengan Java, Dicoding — *Feb 2026*
- Memulai Pemrograman dengan C, Dicoding — *Feb 2026*

COMPETITIONS AND BOOTCAMPS

- QuantifAI Bootcamp, AI and 5D BIM Master Challenge, Glodon Indonesia with the Ministry of Public Works — *Sep 2026*
- Web Design Competition, participant, HIMA-RPL Politeknik Negeri Indramayu — *Apr 2026*

VIRTUAL INTERNSHIP PROGRAMMES

Artificial Intelligence Intern, CodeAlpha *Oct 2026*

One-month remote programme building practical AI assignments and projects.

Python Programming Intern, Oasis Infobyte *Oct 2026*

A month of hands-on Python project work.

C/C++ Development Intern, Codveda Technologies *Feb 2026*

Remote programme covering C and C++ development, testing and optimisation.

Languages. Indonesian (native), English.